

String Manipulation Operations

Q. What is a String in Data Structures? Explain the common string manipulation operations with examples.

> **Definition to Remember**

> In C
refers
all maj

1. String Representation in Memory

char s

2. Common String Manipulation Operations

Function	Operation	Description
----------	-----------	-------------

Function	Operation	Description
----------	-----------	-------------

3. Code Example

```
```c
```

```
#include
```

```
int main() {
```

```
char s
```

```
// 1. Length
```

```
printf("
```

```
// 2. Copy
```

```
strcpy
```

```
// 3. Concatenation
```

```
strcat(
```

```
// 4. Comparison
```

```
if (strcmp
```

```
return 0;
```

```
}
```

```

```

```
> **Must-Write Points (for 10 marks)**
```

```
> 1. A
```

```

```

```
> **Quick Recall**
```

```
> `Strin
```

```

```

## Custom String Manipulation Algorithms & C Codes

### 1. Custom String Operations: `strlen`, `strcpy`, `strcat`, `strrev` (C Code)

```
```c
```

```
#includ
```

```
// Custom strlen
```

```
int my_
```

```
// Custom strcpy
```

```
void m
```

```
// Custom strcat
```

```
void m
```

```
// Custom String Reverse
```

```
void m
```

```
int main() {
```

```
char s
```

```
printf("Length of s1: %d\n", my_strlen(s1));
```

```
my_strcat(s1, s2);
```

```
printf("
```

```
my_strrev(s1);
```

```
printf("
```

```
return 0;
```

```
}
```

```
---
```

2. Naive Pattern Matching Algorithm (C Code)

```
```c
```

```
#includ
```

```
void naivePatternSearch(char text[], char pattern[]) {
```

```
int n =
```

```
for (int i = 0; i <= n - m; i++) {
```

```
int j;
```

```
int main() {
```

char te

```
printf("Text: %s\nPattern: %s\n\n", text, pattern);
```

naiveP

```
return 0;
```

```
}
```